

DNN Theory Notes

Exam Answers

CNN vs ANN, Transfer Learning, Overfitting, BatchNorm, Dropout.

CNN vs ANN

CNN uses local receptive fields and parameter sharing. ANN has many parameters.

Transfer Learning

Pretrained models reduce data and training requirements.

Overfitting

High train accuracy and low test accuracy. Fix with dropout, L2, augmentation.

BatchNorm

Normalizes activations and speeds convergence.

Activation Functions

ReLU, Leaky ReLU, Sigmoid, Softmax formulas.